

Cloud and Mobile Security

Mini Capstone Project

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1. **DNS Service (Amazon Route 53):** A Domain Name System (DNS) service translates human-readable domains (like "www.Instance.Com") into system-readable IP addresses. In the AWS cloud environment, Amazon Route 53 features because the DNS carrier that routes traffic to the internet application.
2. **Web Application Firewall (AWS WAF):** A web utility firewall allows guard net applications from quite a few commonplace attacks, which includes SQL injection and move-site scripting (XSS). In the reference architecture, the web software firewall filters visitors before it reaches the web servers.
3. **Content Delivery Network (Amazon CloudFront):** A content material transport network (CDN) is a geographically dispensed network of servers that can supply content material to customers with lower latency . In the reference structure, Amazon CloudFront acts because the CDN, caching content and serving it to customers from the nearest part region.
4. **Web Servers:** The internet servers host the web software code and deliver content to users [1]. In the reference structure, the internet servers are deployed on Amazon EC2 instances, which might be virtual servers within the AWS cloud.
5. **Auto Scaling Group:** An Auto Scaling group presents a manner to automatically scale internet servers up or down primarily based on site visitors demands. This enables to ensure that the net software has the sources it wishes to address visitors spikes with out compromising overall performance.

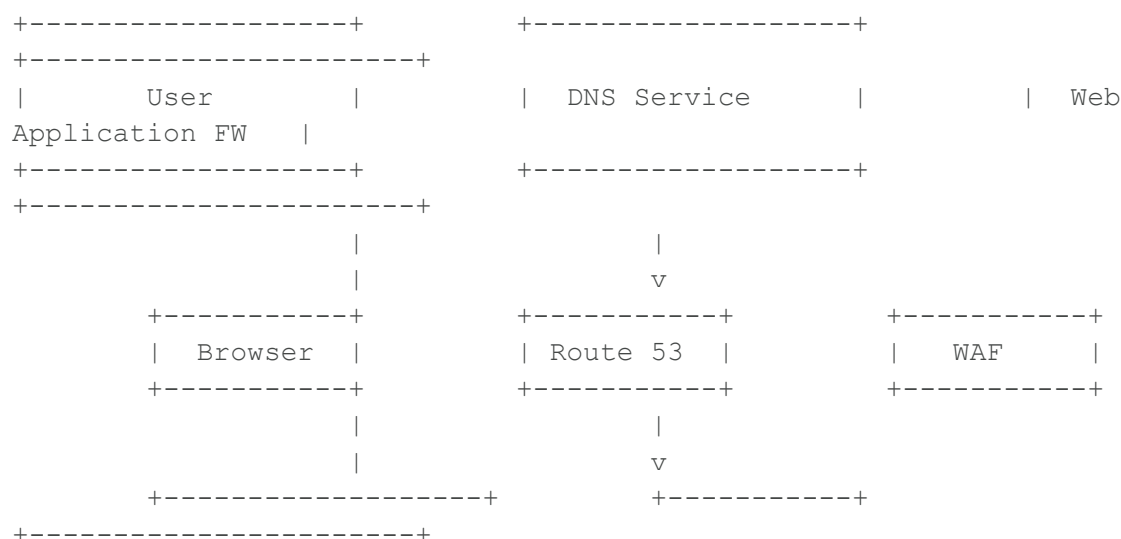
6. **Amazon Simple Storage Service (Amazon S3):** A storage carrier for item garage within the cloud. In the reference structure, Amazon S3 is used for static content material garage and backup functions.
7. **Virtual Private Cloud (VPC):** A virtual non-public cloud (VPC) is a logically remoted segment of the AWS cloud that you may use to release your personal personal network. In the reference architecture, the VPC provides a secure environment for the internet software sources.
8. **Subnet:** A subnet is a segmentation of a VPC that can be used to group assets and define security rules. In the reference structure, the VPC is split into public and private subnets.
 - **Public Subnet:** The public subnet houses resources that want to be accessed from the internet, along with the net servers and the load balancer.
 - **Private Subnet:** The personal subnet homes assets that don't need to be accessed from the internet, along with the database.
9. **Network Address Translation (NAT Gateway):** A NAT gateway allows assets in a non-public subnet to get right of entry to the internet without exposing them to public traffic immediately. In the reference architecture, the NAT gateway permits the web servers in the public subnet to speak with the database in the personal subnet..
10. **Elastic Load Balancing:** A load balancer distributes incoming traffic throughout a couple of net servers in the internet tier. This allows to make sure that no single server is overloaded and that the net application stays exceedingly available.
11. **Amazon Elastic File System (Amazon EFS):** A report device service for storing statistics that needs to be accessed through a couple of EC2 times in a shared way. In the reference structure, Amazon EFS is used for shared garage among the internet servers and the application servers.

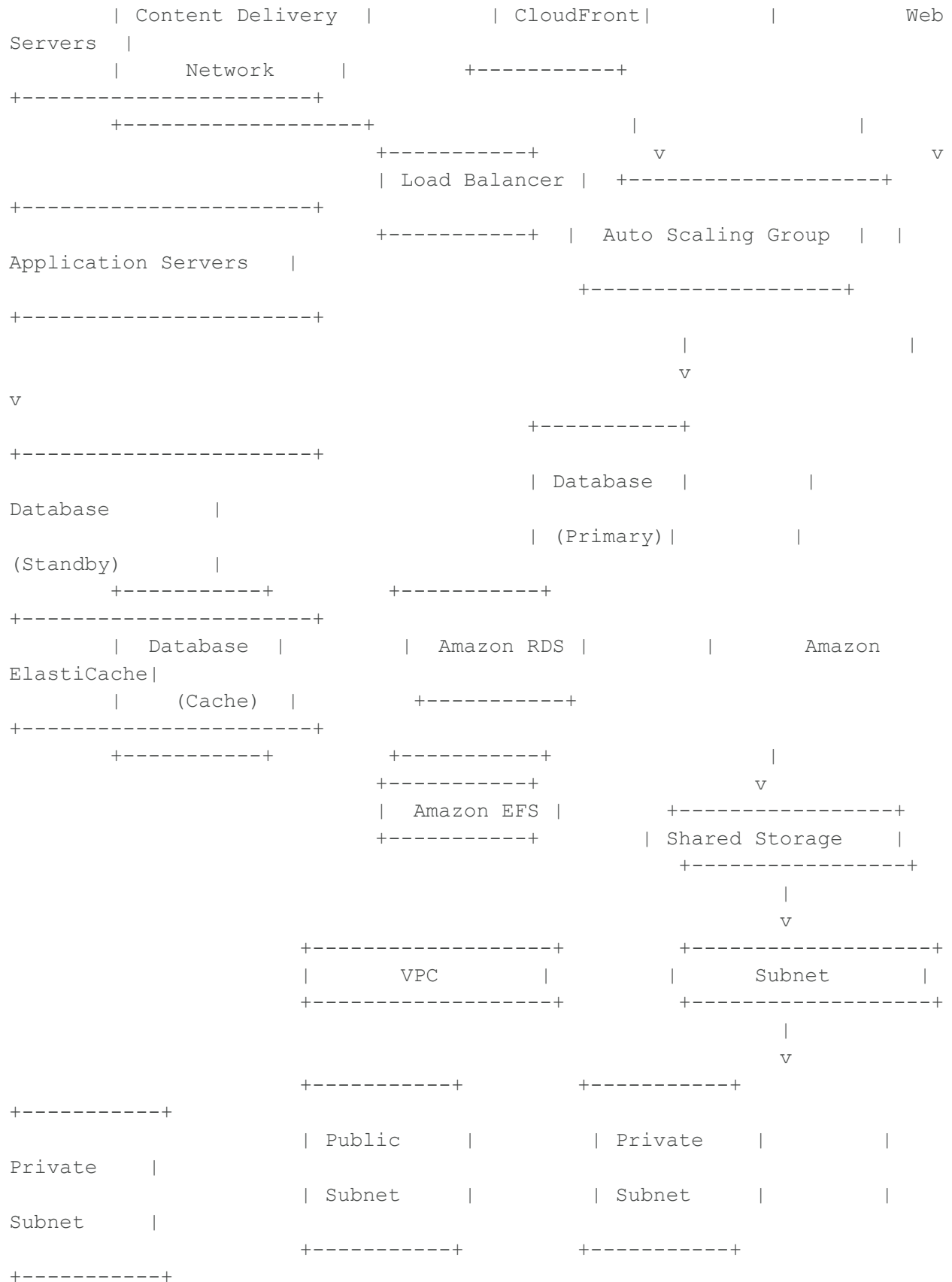
12. **Application Servers:** The application servers host the business good judgment of the web software. In the reference architecture, the software servers are deployed on Amazon EC2 instances in a separate subnet from the net servers.

- Database:** The database stores the web software's facts. In the reference architecture, Amazon Relational Database Service (Amazon RDS) is used for the database.

- **Database (Primary):** The primary database is the main database that serves reads and writes.
 - **Database (Standby):** The standby database is a reproduction of the primary database that can be used for failover in case of an outage.
14. **Amazon ElastiCache:** A caching service which could enhance the overall performance of web applications by means of caching often accessed facts. In the reference architecture, Amazon ElastiCache is used to cache database queries.

Certainly! Here's a flow diagram representing the web application hosting architecture on AWS:





- ### Explanation of the go with the flow:

8. The application servers may have interaction with the database cache (Amazon ElastiCache) for faster facts retrieval.
9. The primary database in Amazon RDS shops and retrieves utility statistics. A standby database offers redundancy in case of outages.
10. The web servers generate the web page content and send the reaction lower back via the burden balancer and CloudFront.
11. Finally, the response reaches the consumer's web browser, displaying the asked internet web page. This go with the flow diagram depicts a high-stage evaluate of the internet utility architecture. The unique configuration would possibly range relying on man or woman wishes.

AWS Migration Proposal Ticket Sheet (CSV Format) :

Ticket ID	Summary	Description	Priority
AWS-MIGR-001	Pre-migration Assessment	Assess current infrastructure and application dependencies.	High
AWS-MIGR-002	Design VPC Architecture	Design VPC architecture and define subnet layout.	High
AWS-MIGR-003	IAM Roles & Policies	Establish IAM roles and policies for users and resources.	High
AWS-MIGR-004	Logging & Monitoring Setup	Configure logging and monitoring using CloudTrail, CloudWatch, and Config.	High
AWS-MIGR-005	Migrate Static Content	Migrate static content (images, CSS, JS) to Amazon S3.	Medium
AWS-MIGR-006	Migrate Database to RDS	Migrate database to Amazon RDS with encryption enabled.	High
AWS-MIGR-007	ElastiCache Configuration	Configure Amazon ElastiCache for caching database queries.	Medium
AWS-MIGR-008	Launch EC2 Instances	Launch EC2 instances for web and application servers.	Medium
AWS-MIGR-009	EC2 Security & IAM	Configure security groups and IAM roles for EC2 instances.	Medium
AWS-MIGR-010	Application Setup	Install necessary software and dependencies on EC2 instances.	Medium
AWS-MIGR-011	Load Balancer Setup	Set up Elastic Load Balancer for distributing traffic.	Medium
AWS-MIGR-012	CloudFront Configuration	Configure CloudFront as CDN for caching and delivery.	Medium
AWS-MIGR-013	Update DNS Records	Update DNS records in Route 53 to point to new AWS resources.	Medium
AWS-MIGR-014	Testing & Optimization	Conduct thorough testing of the migrated application.	High
AWS-MIGR-015	Performance & Security Optimization	Optimize performance and security configurations.	Medium
AWS-MIGR-016	Backup & Disaster Recovery	Implement backup and disaster recovery procedures.	Medium
AWS-MIGR-017	Post-Migration Monitoring	Monitor application performance and security using CloudWatch and CloudTrail.	Medium
AWS-MIGR-018	Staff Training & Support	Provide support and training to operational staff.	Medium
AWS-MIGR-019	Document Architecture	Document architecture and configurations for future reference.	Medium

Diagram :

